

8.1 Stationary Waves

Question Paper

Course	CIEA Level Physics
Section	8. Superposition
Topic	8.1 Stationary Waves
Difficulty	Medium

Time allowed: 40

Score: /27

Percentage: /100

Question 1

(i)

By reference to the direction of transfer of energy, state what is meant by a transverse wave.

[1]

(ii)

State the principle of superposition.

[2]

[3 marks]

Question 2a

A stationary wave on a string is shown in Fig. 1.1.

Explain how the wave is formed, referring to the principle of superposition in your answer.

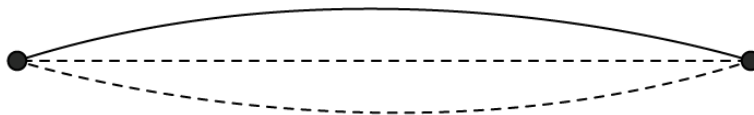


Fig. 1.1

[4 marks]

Question 2b

On Fig. 1.1, draw the stationary wave that would be formed on the string in part (a) with two more nodes and two more antinodes.

[1 mark]

Question 2c

Fig. 1.2 shows the appearance of a stationary wave on a stretched string at one instant in time.

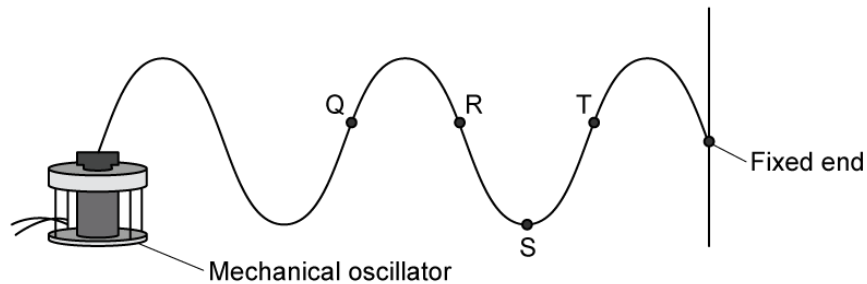


Fig. 1.2

Mark clearly on the diagram

- (i)
the equilibrium position,
- (ii)
at least two nodes by labelling them **N**, and two antinodes by labelling them **A**,
- (iii)
the direction in which points **Q**, **R**, **S** and **T** are about to move.

[1]

[2]

[2]

[5 marks]

Question 2d

For the wave in Fig. 1.2 the frequency of vibration is 180 Hz and the speed of the waves along the string is 60 m s^{-1} .

For this wave

(i)

Calculate the time period the stationary wave on the string.

[1]

(ii)

Calculate the length of the string.

[2]

[3 marks]

Question 3a

Fig. 1.1 shows a stationary wave formed on a guitar string fixed at **P** and **Q** when it is plucked at its centre.

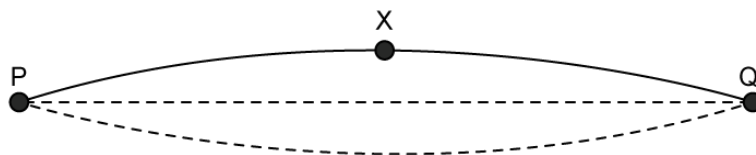


Fig. 1.1

X is a point on the string at maximum displacement.

Explain why a stationary wave is formed on the string

[3 marks]

Question 3b

The stationary wave in Fig. 1.1 is the D string of the guitar which has a frequency of 146.83 Hz.

Calculate the time taken for the string at point X to move from maximum displacement to its next maximum displacement.

[3 marks]

Question 3c

The progressive waves on the string travel at a speed of 190 m s^{-1} .

Calculate length of the D string.

[2 marks]

Question 3d

A guitarist presses on the string at point R to shorten it and create the higher note 'E'. The distance between R and Q in Fig. 1.2 is 0.29 m.

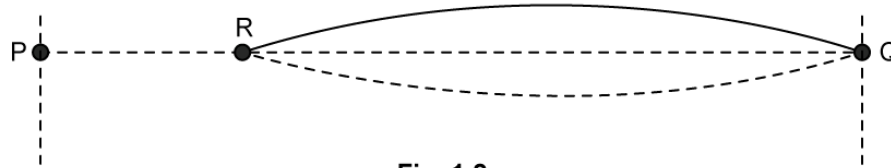


Fig. 1.2

The speed of the progressive wave remains at 190 m s^{-1} and the tension remains constant.

Calculate the frequency of note E.

[3 marks]



Model Answers: Medium

1

(i) A transverse wave is defined as:

- (Particle) vibrations / oscillations that are perpendicular to the direction of energy transfer; [1 mark]

(ii) The principle of superposition states that:

- Waves meet / overlap at a point; [1 mark]
- The resultant displacement is the sum of the individual displacements; [1 mark]

[Total: 3 marks]

2a

(a) Explain how the wave is formed, referring to the principle of superposition in your answer:

Explain how the wave is formed:

- (Progressive waves) travel (from the centre) to the end(s) and then reflect; [1 mark]
- Two (progressive) waves travel in opposite directions along the string having the same/similar frequency/wavelength/amplitude; [1 mark]

Use the principle of superposition:

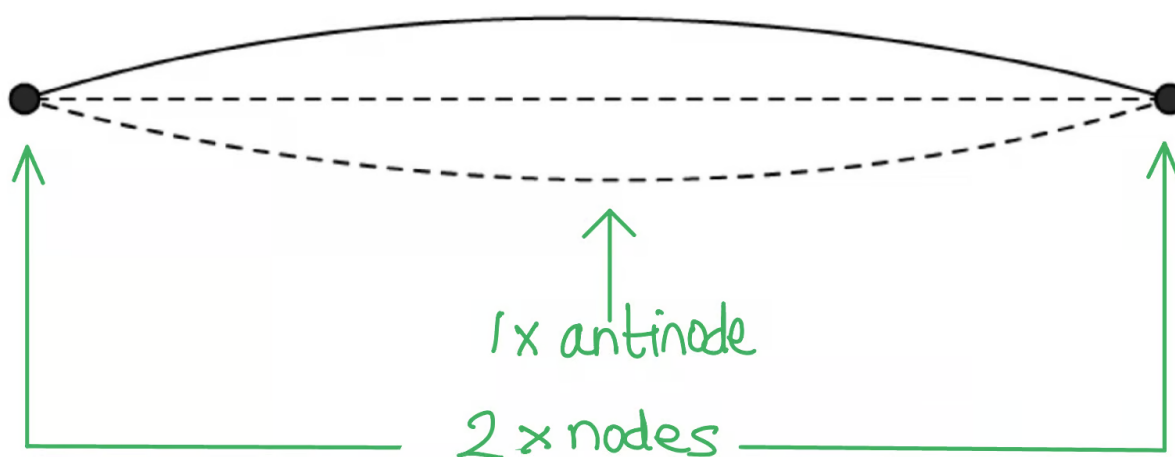
- (The two waves) superpose/interfere to produce the stationary wave; [1 mark]
- The (maximum) displacement is twice the initial maximum displacement / equal to the sum of the displacements of the individual waves; [1 mark]

[Total: 4 marks]

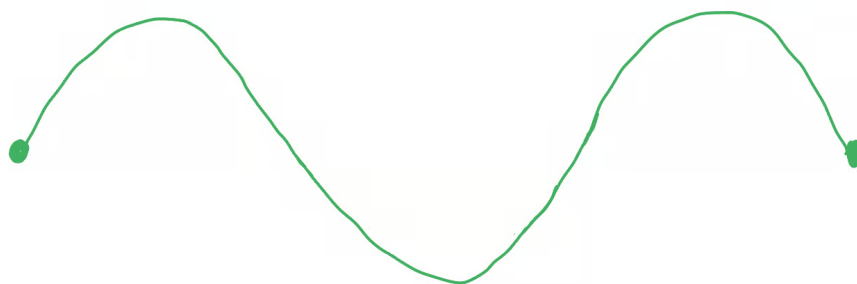
2b

(b) On Fig. 1.1, draw the stationary wave that would be formed on the string in part (a) with two more nodes and two more antinodes.

Count the nodes and antinodes:



Draw a stationary wave with 4 nodes and three antinodes:

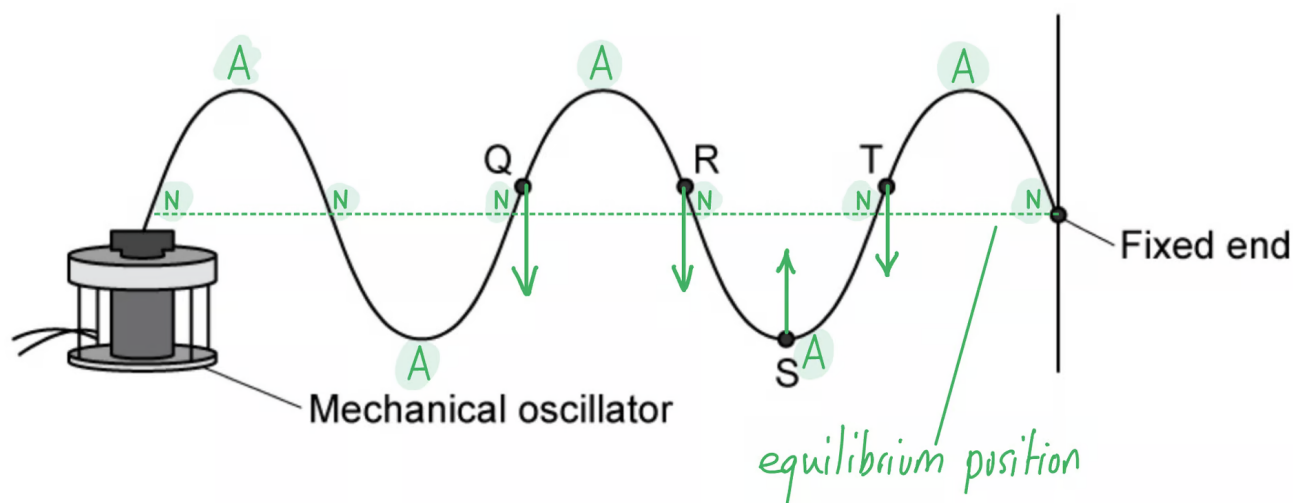


- Diagram as shown above; [1 mark]

[Total: 1 mark]

2c

(c) A correctly annotated diagram looks like:



(i) Mark clearly on the diagram the equilibrium position

- Equilibrium position horizontal and meets fixed end; [1 mark]

(ii) Mark clearly on the diagram at least two nodes by labelling them **N**, and two antinodes by labelling them **A**

- At least two correct **N** [1 mark]
- At least two correct **A** [1 mark]

(iii) Mark clearly on the diagram the direction in which points **Q**, **R**, **S** and **T** are about to move.

- **Q**, **R** and **T** moving downwards; [1 mark]
- **S** moving upwards; [1 mark]

[Total: 5 marks]

2d

(d)

(i) Calculate the time period the stationary wave on the string:

Use the equation relating frequency and time period:

- $T = \frac{1}{f} = \frac{1}{180} = 5.6 \times 10^{-3} \text{ s or } 5.6 \text{ ms}$ [1 mark]

(ii) Calculate the length of the string:

List the known quantities:

- Frequency, $f = 180 \text{ Hz}$
- Speed, $v = 60 \text{ m s}^{-1}$
- Number of complete wavelengths = 2.5 (counted from the diagram)

Use the wave equation to find wavelength:

- $v = f \lambda$
- $\lambda = \frac{v}{f} = \frac{60}{180} = 0.33 \text{ m [1 mark]}$

Multiply λ by the number of wavelengths:

- Length = $2.5 \times 0.33 = 0.83 \text{ m}$ or 83 cm [1 mark]

[Total: 3 marks]

3a

(a) Explain why a stationary wave is formed on the string:

Any **three** from:

- The progressive waves of the string travel (from the centre) to the ends/points P and Q and reflect; [1 mark]
- Two (progressive) waves travel in opposite directions along the string; [1 mark]
- (The two waves have) the same frequency/wavelength; [1 mark]
- (The two waves have) the same/similar amplitude; [1 mark]
- (The two waves) superpose/interfere to produce the stationary wave; [1 mark]

[Total: 3 marks]

3b

(b) Calculate the time taken for the string at point X to move from maximum displacement to its next maximum displacement:

List the known quantities:

- Frequency, $f = 146.83 \text{ Hz}$

Determine how much time elapses between maxima:



- Length = $2.5 \times 0.33 = 0.83 \text{ m}$ or 83 cm [1 mark]

[Total: 3 marks]

3a

(a) Explain why a stationary wave is formed on the string:

Any **three** from:

- The progressive waves of the string travel (from the centre) to the ends/points P and Q and reflect; [1 mark]
- Two (progressive) waves travel in opposite directions along the string; [1 mark]
- (The two waves have) the same frequency/wavelength; [1 mark]
- (The two waves have) the same/similar amplitude; [1 mark]
- (The two waves) superpose/interfere to produce the stationary wave; [1 mark]

[Total: 3 marks]

3b

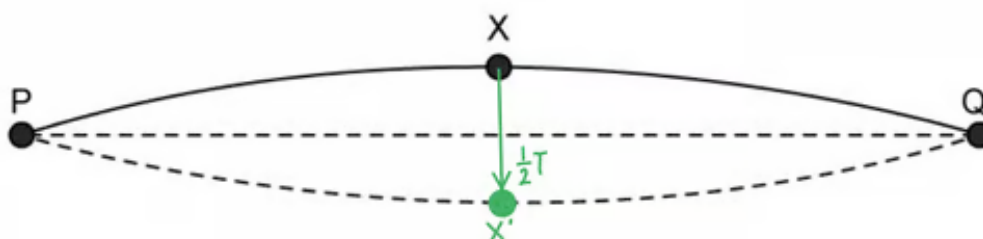
(b) Calculate the time taken for the string at point X to move from maximum displacement to its next maximum displacement:

List the known quantities:

- Frequency, $f = 146.83 \text{ Hz}$

Determine how much time elapses between maxima:

- The next maximum displacement is after $\frac{1}{2}$ a cycle; [1 mark]



Calculate the time taken:

- Time period for one cycle, $T = \frac{1}{f} = \frac{1}{146.83} = 6.8 \times 10^{-3} \text{ s}$ [1 mark]
- Time period for $\frac{1}{2}$ a cycle $= 0.5 \times (6.8 \times 10^{-3}) = 3.4 \times 10^{-3} \text{ s}$ [1 mark]

[Total: 3 marks]

3c

(c) Calculate length of the D string:

List the known quantities:

- Frequency, $f = 146.83 \text{ Hz}$
- Speed, $v = 190 \text{ m s}^{-1}$

Determine the wavelength using the wave equation:

- $v = f \lambda$
- $\lambda = \frac{v}{f} = \frac{190}{146.83} = 1.29 \text{ m}$ [1 mark]

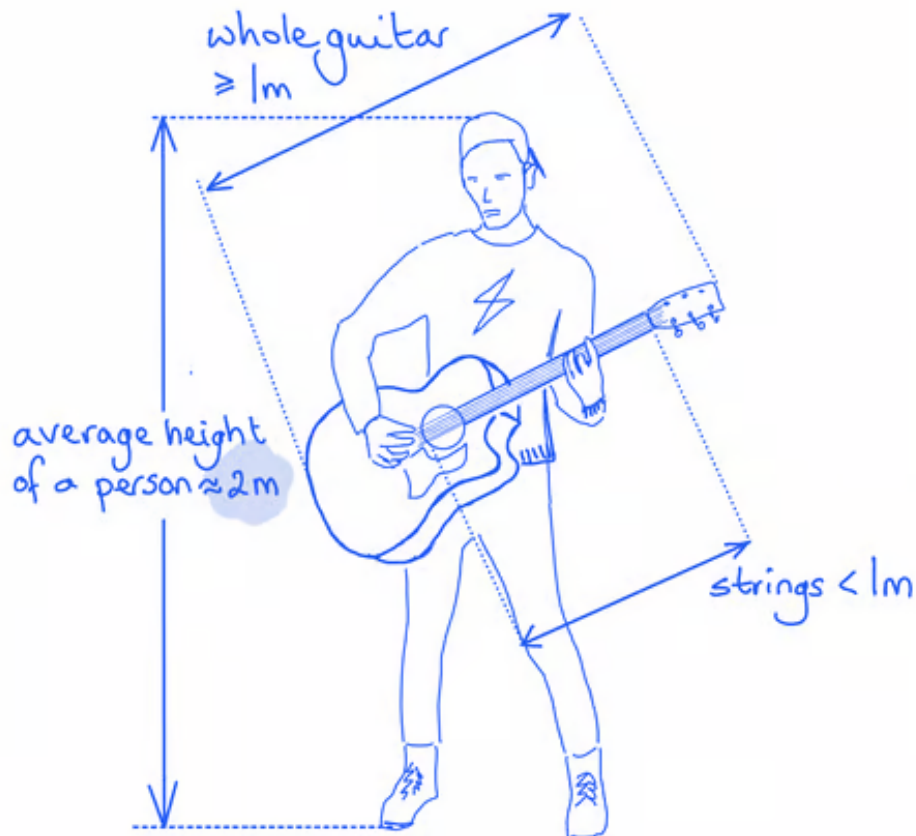
Count the number of wavelengths and calculate the length of the string:

- Fig 1.1 shows half a wavelength
- Length of string, $L = 0.5 \times 1.29 = 0.65 \text{ m}$ or 65 cm [1 mark]

[Total: 2 marks]

In a question like this one you can use a certain amount of common sense. You may not play guitar, but you have some idea of how large they are.

A person holding a guitar is nearly twice as long as the whole instrument, so a guitar must be about a metre long. The strings take up much less than the full length, so they must be less than one metre.



Picturing a guitar player in this way shows that the strings are very likely much less than a meter long, not quite a lot more. This should remind you to halve the wavelength. Try to become confident estimating real-life values by practicing visualisation, and even by measuring things to check your estimate!

3d

(d) Calculate the frequency of note E:

List the known quantities:

- Speed of the wave, $v = 190 \text{ m s}^{-1}$
- Distance RQ = 0.29 m

Determine the wavelength:

- wavelength = $2 \times \text{length of string (RQ)}$ [1 mark]
- $\lambda = 2 \times 0.29 = 0.58 \text{ m}$ [1 mark]

Frequency calculation:

- $f = \frac{v}{\lambda} = \frac{190}{0.58} = 327.586 = 330 \text{ Hz [1 mark]}$